DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	**ITCP322**
COURSE TITLE	**:**	**Capstone Project 1**
COURSE TYPE	**:**	Lecture / Research
COURSE CREDIT	**:**	Three Units
LEARNING MODALITY	**:**	Traditional Blended Learning Model
PRE-REQUISITES	**:**	Should be 3rd year level
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Paulino H. Gatpandan, DIT
CONSULTATION HOURS	**:**	0800 – 1000 Monday (COS100C), Thursday (Online) 1130 – 1200 Monday (COS100C), Thursday (Online) 1300 – 1430 Monday (COS100C), Thursday (Online) 1600 – 1700 Monday (COS100C) 1100 – 1200 Wednesday (Online)

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

A project focusing on the information system plan brought about in the practicum / on-the-job training / independent research course. This course requires the fulfillment of the complete documentation and oral defense before a set of panelists.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Understand the latest concepts in systems analysis and design, software engineering and information technology.
CLO2. Investigate and evaluate prominent literature connected to your Capstone Project.

CLO3. Understand the scope of the project, planning, scheduling the collaborative efforts of the team, and documenting the effort via oral and written proposals.
CLO4. Illustrate that they can function, communicate, collaborate and continue to learn effectively as ethically and socially responsible IT professionals.
CLO5. Adapt the skills to communicate with and balance the interests of multiple stakeholders (both technical and non-technical) on a project.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand the latest concepts in systems analysis and design, software engineering and information technology.	TLO1. Identify complex problems and define the computing requirements appropriate to its solution. TLO2. Identify the user needs and consider them in the selection, creation, and evaluation and administration of computer-based systems. TLO3. Select various high level and detailed design and implementation principles for building a software system.
CLO2. Investigate and evaluate prominent literature connected to your Capstone Project	TLO4. Analyze the impact of computing and engineering solutions on individuals, organizations, and society.
CLO3. Understand the scope of the project, planning, scheduling the collaborative efforts of the team, and documenting the effort via oral and written proposals.	TLO5. Identify software development needs and challenges that require various engineering solutions, and formulate such solutions
CLO4. Illustrate that they can function, communicate, collaborate and continue to learn effectively as ethically and socially responsible IT professionals.	TLO6. Execute effectively as a member or leader of a development team recognizing the different roles of a team to accomplish a common goal. TLO7. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
CLO5. Adapt the skills to communicate with and balance the interests of multiple stakeholders (both technical and non-technical) on a project.	TLO8. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
CLO6. Demonstrate 21st century skills in all learning activities.	TLO9. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Introduction Module	Week 1 Jan 20-25	**Course Orientation** • Gospel Reading, Reflection and Prayer • Course Overview • Course Polices and Requirements • Professor Profile • Grading System • Netiquette • Undergraduate Research Manual and the Capstone Project Guidelines Discussion	Schoolbook MS Teams Power Point	1.5	1.5	3.0
Module 1 CLO1 CLO2 CLO6 TLO1 TLO2 TLO3 TLO4 TLO9	Week 2 Jan 27 – Feb 1	**Introduction to Research** • Gospel Reading, Reflection and Prayer • Introduction to Research • Classification and Characteristics of Research	Schoolbook MS Teams Power Point	1.5		3.0
		• **Enabling assessment:** Submission of 3 Research Titles of Capstone Project	Schoolbook		1.5	
Module 2 CLO1 CLO2 CLO3 CLO6 TLO1 TLO2 TLO3 TLO4 TLO5	Week 3 Feb 3-8	**Chapter 1 - Introduction** • Gospel Reading, Reflection and Prayer • Background of the Study • The Research Problem • Statement of Objectives • Significance of the Study • Scope and Limitations of the Study • Endorsement to Advisers	Schoolbook MS Teams Power Point	1.5	1.5	3.0

Research Work CLO1 CLO6 TLO1 TLO2 TLO3 TLO9	Week 4 Feb 10-15	• Research work (Chapter 1) • Consultation with Adviser **Enabling assessment:** Submission of Chapter 1 of the proposal (first choice)	Schoolbook		3.0	3.0
Module 3 CLO2 CLO6 TLO4 TLO9	Week 5 Feb 17-22	**Chapter 2 - Review of Related Literature** • Gospel Reading, Reflection and Prayer • Conceptual Framework • Synthesis	Schoolbook MS Teams Power Point	1.5	1.5	3.0
Research Work CLO2 CLO6 TLO4 TLO9	Week 6-7 Feb 24 - Mar 8	• Research work (Chapter 2) • Consultation with Adviser **Enabling assessment:** Submission of Chapter 2	Schoolbook		6.0	6.0
Module 4 CLO1 CLO2 CLO3 CLO6 TLO1 TLO2 TLO3 TLO4 TLO5	Week 8 Mar 10-15	**Chapter 3 - Methodology** • Gospel Reading, Reflection and Prayer • Research Methodology • Research Instruments • Software Methodology	Schoolbook MS Teams Power Point	1.5	1.5	3.0
	Week 9 Mar 17-22	**Summative Assessment:** Written examination on all chapters discussed	Schoolbook	1.5	1.5	3.0
			Total	**9**	**18**	**27**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand the latest concepts in systems analysis and design, software engineering and information technology.	TLO1. Identify complex problems and define the computing requirements appropriate to its solution. TLO2. Identify the user needs and consider them in the selection, creation, and evaluation and administration of computer-based systems.

	TLO3. Select various high level and detailed design and implementation principles for building a software system.
CLO2. Investigate and evaluate prominent literature connected to your Capstone Project	TLO4. Analyze the impact of computing and engineering solutions on individuals, organizations, and society.
CLO3. Understand the scope of the project, planning, scheduling the collaborative efforts of the team, and documenting the effort via oral and written proposals.	TLO5. Identify software development needs and challenges that require various engineering solutions, and formulate such solutions
CLO4. Illustrate that they can function, communicate, collaborate and continue to learn effectively as ethically and socially responsible IT professionals.	TLO6. Execute effectively as a member or leader of a development team recognizing the different roles of a team to accomplish a common goal. TLO7. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
CLO5. Adapt the skills to communicate with and balance the interests of multiple stakeholders (both technical and non-technical) on a project.	TLO8. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
CLO6. Demonstrate 21st century skills in all learning activities.	TLO9. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	**Date**	**Teaching-Learning Lessons and Activities / Assessment Strategies**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**
Module 5 CLO4 TLO7	Week 10 Mar 24-29	**Intellectual Property** • Gospel Reading, Reflection and Prayer • Ethical Issues in Research Intellectual Property Rights	Schoolbook MS Teams Power Point	1.5	1.5	3.0

Research Work CLO3 CLO6 TLO5 TLO9	Week 11-12 Mar 31 - Apr 12	• Gospel Reading, Reflection and Prayer • Consultation with Advisers	Schoolbook		6.0	6.0
	Week 13 Apr 14-19	Independent Learning Days / Holy Week	Schoolbook		3.0	3.0
Research Work CLO3 CLO4 CLO6 TLO5 TLO6 TLO7 TLO9	Week 14 Apr 21-26	• Manuscript Writing • Consultation with Advisers **Enabling assessment:** Submission of Progress Report and Endorsement Form	Schoolbook		3.0	3.0
Proposal Defense CLO3 CLO4 CLO5 CLO6 TLO5 TLO6 TLO7 TLO8 TLO9	Week 15-18 Apr 28 - May 24	**Summative Assessment**: Oral Presentation (Proposal) • Gospel Reading, Reflection and Prayer	MS Teams	12		12.0
			Total	**13.5**	**13.5**	**27**

GRADING SYSTEM:

Midterm

Enabling Assessments	- 30%
Summative Assessments	- 70%
	100%

Finals

Enabling Assessments (Research Teacher – 10%, Adviser – 20%)	- 30%
Summative Assessments (RRP Members Grade)	- 70%
	100%

Overall Final Grade = Midterm * 0.40 + Final *0.60

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Call number or e-provider	**References**
	Undergraduate Research Handbook: Policies and Guidelines
QA 402 .D423a 2019	Dennis A, Wixom B, & Roth R. 2019. Systems analysis and design. 7th edition. Hoboken, NJ : Wiley
QA 76.9.D26 .C816 2019	Coronel C & Morris S. 2019. Database systems: design, implementation, and management. 13th edition. Australia; United States : Cengage Learning
eBooks	Ariola M. 2006. Principles and methods of research. Rex Bookstore Inc.
Journals	Z. Gao, M. P. Gatpandan and P. H. Gatpandan, "Classification Decision Tree Algorithm in Predicting Students' Course Preference," *2021 2nd International Symposium on Computer Engineering and Intelligent Communications (ISCEIC)*, Nanjing, China, 2021, pp. 93-97, doi: 10.1109/ISCEIC53685.2021.00026. J. Xu, P. H. Gatpandan, M. P. Gatpandan and Z. Gao, "Freight and Passenger Volume Prediction using Multiple Regression," *2021 International Conference on Machine Learning and Intelligent Systems Engineering (MLISE)*, Chongqing, China, 2021, pp. 77-83, doi: 10.1109/MLISE54096.2021.00022. X. Gao, M. F. Rosas, Y. Zhang, M. P. Gatpandan, P. H. Gatpandan and D. Dasig, "The Research and Analysis of Student Health Status on the Basis of Physic Fitness Test' Data," *2021 International Conference on Public Health and Data Science (ICPHDS)*, Chengdu, China, 2021, pp. 110-115, doi: 10.1109/ICPHDS53608.2021.00031. D. J. R. Calantoc, D. D. Dasig, E. E. Claricia, M. A. B. Taduyo, P. H. Gatpandan and R. V. F. Guarin, "Clustering and Predicting Smartphones Features using Gaussian Mixture Model Algorithm," *2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, Manila, Philippines, 2021, pp. 1-6, doi: 10.1109/HNICEM54116.2021.9731964.

Prepared: **Information Technology Department**
2nd Semester 2024-2025

Approved: **(SGD)Ms. Sherry B. Naz**
Chair, Information Technology Department

Endorsed: **(SGD)Marivic R. Mitschek**
Dean, College of Information and Computer Studies

cf: Office of the Provost, AEA, CICS Curriculum Development Committee, file